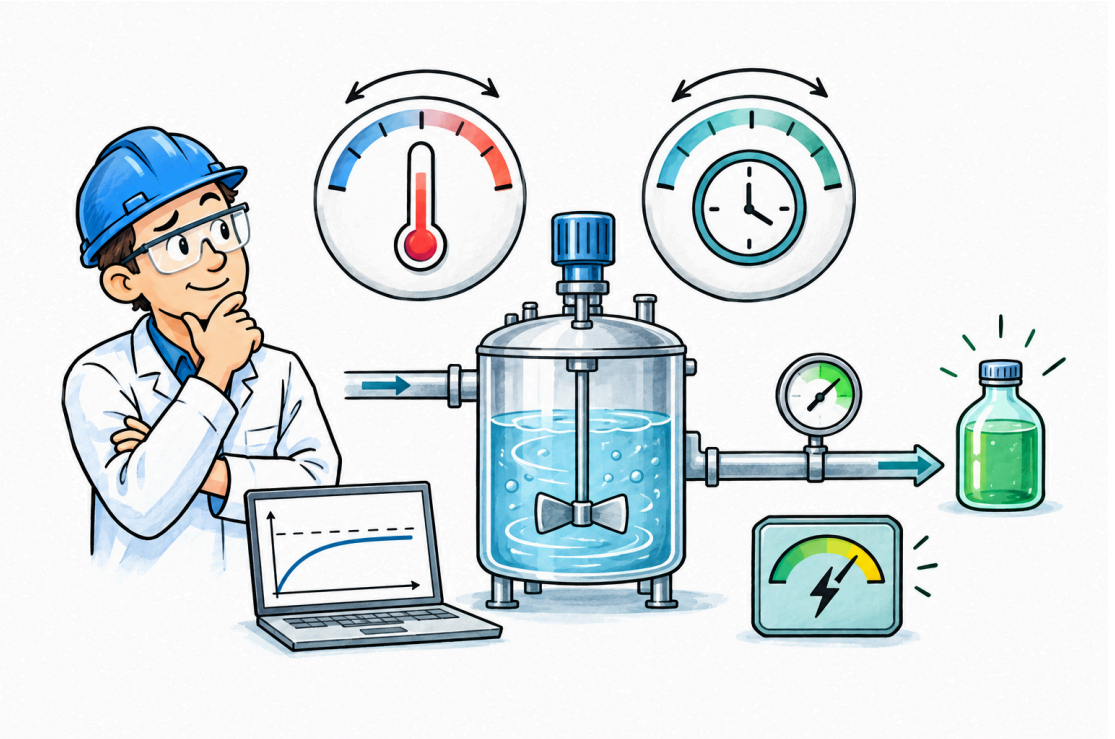
Week 1: Process Prediction and Surrogate Modeling Basics

Outlet concentration prediction at specified operating conditions

An engineer wants to predict outlet concentration and conversion at new temperature and residence-time settings.

Experiments or detailed process calculations take time at each condition. Could existing data help predict the response?



Prediction performance at new operating conditions

A model can reproduce its training data and still make large errors at new inputs. Units, operating range, and fixed assumptions also determine what its output means.

This week we define a prediction task and evaluate it on independent data. A small CSTR with a computable reference response lets us check each label.

Predictor fitting and independent evaluation

1. Define inputs, outputs, context, units, and the domain.
2. Generate steady-state reference data and keep train, validation, and test separate.
3. Fit a quadratic regression baseline and report test errors in physical units.
4. Inspect concentration-conversion consistency and where the errors occur.

Course roadmap: Week 2 learns multicomponent MLPs; Week 3 adds time response; Week 4 examines physical consistency. Weeks 5-7 address optimization formulations and model structure. Week 8 selects and revalidates operating conditions.

Submit a predictor with an independent evaluation.

1. Define the prediction contract

Role	This example
Inputs	Temperature T [K], residence time τ [min], feed concentration C_Af [mol/L]
Outputs	Outlet concentration C_A [mol/L], conversion X [−]
Domain	300–360 K; 1–10 min; 0.8–1.5 mol/L
Fixed context	Ideal mixing, constant volume/density, first-order A → B, declared rate parameters
Reference	Steady-state solution of the specified CSTR balance

Feed concentration varies in this prediction task.

2. Reference CSTR model

Use an ideal, perfectly mixed, constant-volume, isothermal CSTR with a first-order reaction A → B, constant density, and no B in the feed. Reactor temperature T is an operating input.

$$\frac{dC_A}{dt} = \frac{C_{Af} - C_A}{\tau} - k\left(T\right)C_A$$

Here concentrations are in mol/L, time and residence time in minutes, T in kelvin, and k in inverse minutes. The flow-to-volume ratio is the reciprocal of residence time. The inlet-outlet term and the reaction term therefore both have units of mol/(L·min).

Temperature dependence and operating domain

The rate constant depends on temperature as follows:

$$k\left(T\right) = k_{\mathrm{ref}}\exp\left[\beta\left(\frac{1}{T_{\mathrm{ref}}} - \frac{1}{T}\right)\right]$$

Parameter or input	Value or range	Interpretation
Reference rate constant	0.2 per minute	Rate at the reference temperature
Reference temperature	330 K	Rate-law reference
β = E/R	6000 K	Synthetic temperature-sensitivity parameter
Temperature	300–360 K	Sampling and evaluation domain
Residence time	1–10 min	Sampling and evaluation domain
Feed A concentration	0.8–1.5 mol/L	Sampling domain; reference map at 1.2

3. Derive the steady-state map

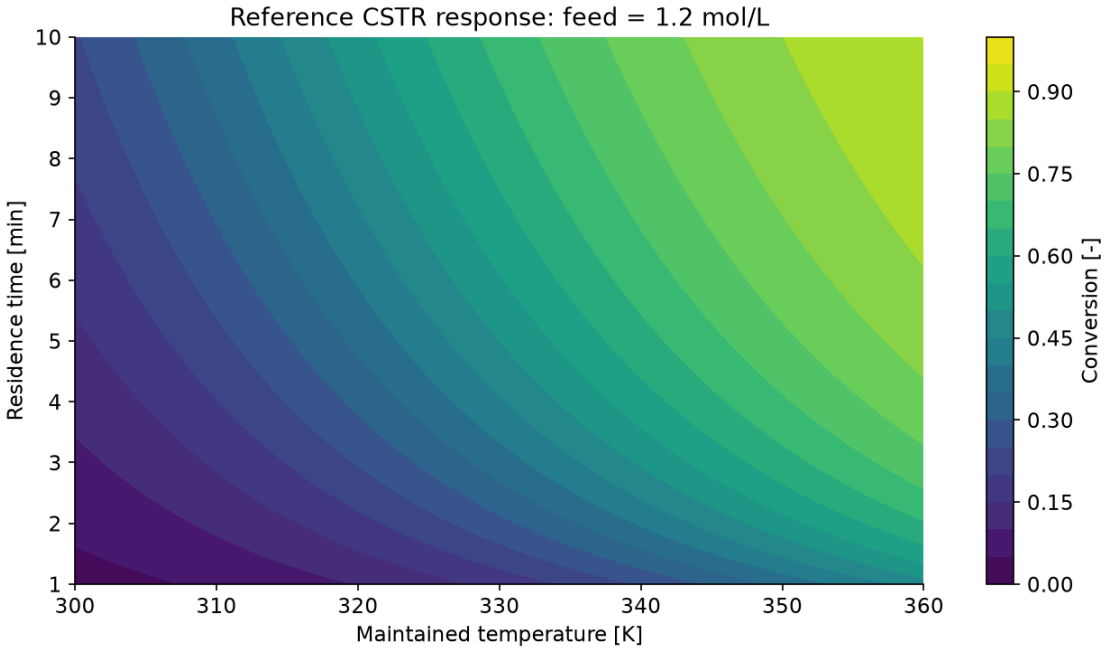
Set the time derivative to zero, multiply by residence time, and collect the terms containing outlet concentration. The resulting mapping and conversion are:

$$C_A = \frac{C_{Af}}{1 + k(T)\tau}, \quad X = 1 - \frac{C_A}{C_{Af}} = \frac{k(T)\tau}{1 + k(T)\tau}$$

This gives a three-input, two-output map: temperature, residence time, and feed concentration map to outlet concentration and conversion. In this first-order model, conversion is independent of feed concentration.

Read the reference response surface

At 330 K and 5 min, conversion is 0.5; with feed 1.2 mol/L, outlet concentration is 0.6 mol/L.



4. Keep data roles separate

Generate 400 train, 120 validation, and 160 test points independently inside the declared box, with seed 42.

Fit coefficients with train data. Use validation for model choices. Reserve test for the final error report. The supplied baseline has a fixed quadratic basis.

5. Fit a small quadratic baseline

Scale each input using its declared lower and upper bounds:

$$\tilde{\mathrm{x}}_j = 2\frac{x_j - l_j}{u_j - l_j} - 1$$

Use ten features: a constant, three scaled inputs, three squares, and three pairwise products. The feature matrix Φ is 400 × 10; Y is 400 × 2; B is 10 × 2.

Fit both outputs by least squares:

$$\min_B \|\Phi B - Y\|_F^2, \quad \hat{y} = \varphi\left(x\right)B$$

At prediction time, keep the fitted regression coefficients fixed and evaluate a new input.

6. Evaluate errors and output consistency

Evaluate each output on the held-out test set:

$$\mathrm{RMSE}_j = \sqrt{\frac{1}{N}\sum_{n=1}^{N}\left(\hat{y}_{nj} - y_{nj}\right)^2}$$

Concentration RMSE is in mol/L. Conversion RMSE is dimensionless; 0.025608 corresponds to about 2.56 percentage points. Also check whether the predicted pair agrees with the definition of conversion.

$$r = \hat{C_A} + C_{\mathrm{Af}}\hat{X} - C_{\mathrm{Af}}$$

The reference gives r = 0. An unconstrained two-output regression can disagree. A small average RMSE does not mean every condition has a small error.

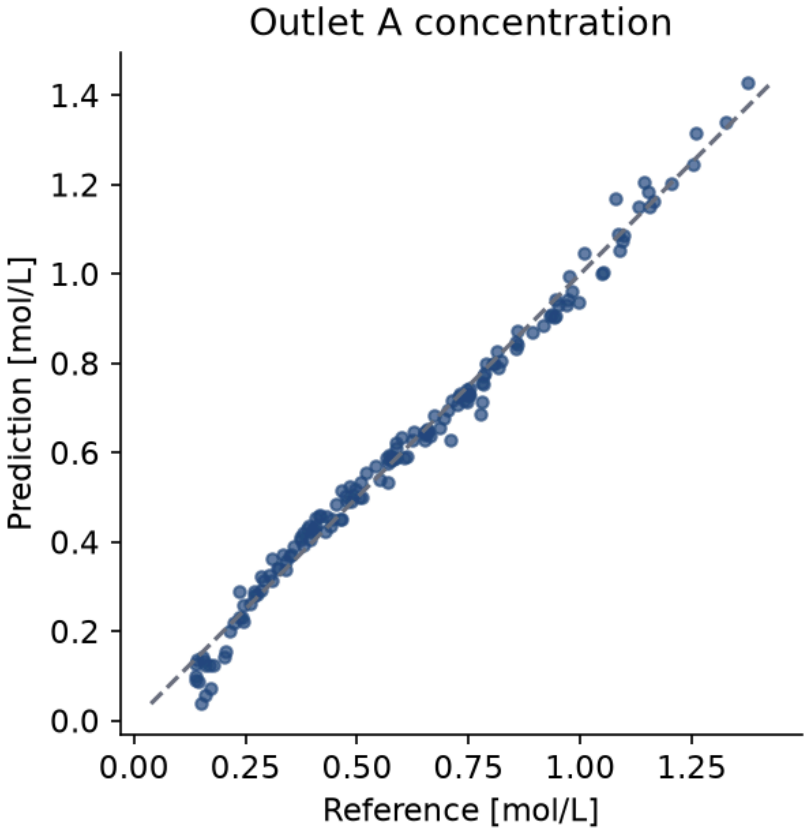
Prediction results to reproduce

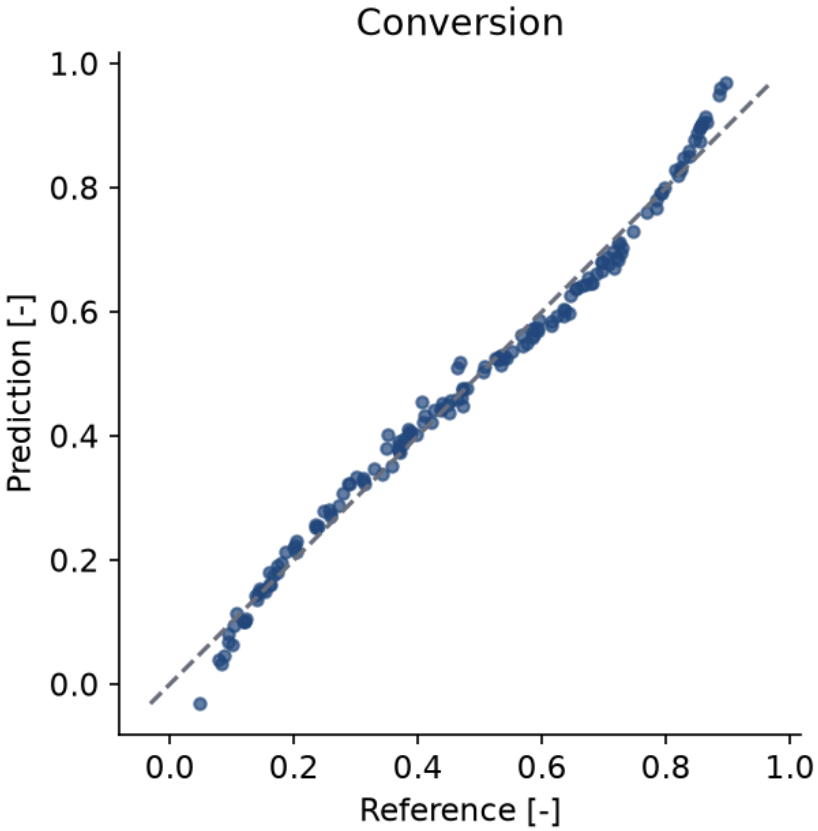
Check	Seed-42 baseline
Test concentration RMSE	0.032623 mol/L
Test conversion RMSE	0.025608
Test samples	160 independent points
Model	Least squares on ten quadratic features

Judge whether these errors are small enough for a particular use by inspecting parity plots and residuals together.

Prediction error across operating conditions

The diagonal represents prediction equal to reference. Compare concentration and conversion separately, then relate the larger errors to the input conditions.





7. Run the prediction lab

Download the notebook or script, plus requirements.txt.

```
python -m pip install -r requirements.txt
python week01_lab.py --output-dir week01-results --seed 42
```

The default run saves the dataset, prediction metrics, the reference map, and test parity plots. Compare the published dataset and recorded results.

Optional later-topic examples are retained in the script behind `--extensions`.

8. Exercises and required submission

1. Derive C_A and X from the steady-state balance, with units.
2. Calculate the response at 330 K, 10 min, and feed 1.2 mol/L.
3. Count the quadratic features and state the dimensions of Φ, B, and Y.
4. Re-run with another seed and compare held-out errors. Explain why train error alone is insufficient.
5. For predicted $C_A = 0.65$ mol/L and $X = 0.45$ at feed 1.2 mol/L, calculate the consistency residual.

Submit a prediction specification, the fitted baseline, and a test report with units, parity plots, and consistency diagnostics.

Answer checks: $C_A = 0.4$ mol/L and $X = 2/3$ in Question 2; ten features; residual -0.01 mol/L in Question 5.

Reading and the next prediction task

Work through the data definition, CSTR derivation, input scaling, regression, and independent evaluation in the Week 1 companion or English PDF, then check your answers.

Using a predictor to select operating conditions requires objectives, operating constraints, and reference-model revalidation. Weeks 5-8 develop that process.